MINI PROJECT

PROMPT DOCUMENTATION

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Course: B.Tech – Computer Science and Engineering

University: ITM Skills University

Subject: AI Tools & Prompt Engineering

FormFlow AI – AI-Powered Form Generation and Validation Assistant

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1. Introduction

Prompt engineering is the process of designing and improving instructions given to an AI tool to obtain accurate and useful results.

For this project, ChatGPT was used as an assistance tool to plan, develop, improve and troubleshoot a web-based application called **FormFlow AI**.

The project focuses on simplifying form creation. Users can enter a text prompt describing the form they need, generate a form, validate its fields, download completed details as a PDF and share a link with others.

The project was developed through multiple instructions and improvements rather than using only one prompt.

2. Project Objective

The main objective was to develop a simple and user-friendly form generation and validation assistant.

The project aims to:

- Generate forms based on user-provided prompts.
- Identify suitable form fields and input types.
- Provide an interactive form preview.
- Validate user inputs.
- Allow completed forms to be downloaded as PDFs.
- Generate shareable links that allow other users to access blank forms.

3. Tools and Technologies Used

Tool / Technology	Purpose
ChatGPT	Planning, code generation, explanations and troubleshooting
HTML	Creating the website structure

CSS	Designing the interface and layout
JavaScript	Form generation, validation and interactions
GitHub	Managing and storing project code
Vercel	Deploying the website online
Gamma AI	Creating the project presentation

4. Prompt Engineering Process

The project was developed by starting with a broad requirement and gradually adding more specific instructions.

Each stage improved the clarity, functionality or design of the application.

Prompt 1 – Defining the Project

Initial Prompt:

"Help me design and develop an AI-powered form generation and validation assistant. The application should allow users to enter prompts and automatically generate forms based on their requirements."

Purpose:

The first prompt was used to understand the project idea and identify its main purpose.

Improvement:

The initial idea was broad, so additional requirements were introduced to define the application's appearance and functionality.

Prompt 2 – Improving the Website Design

Prompt:

"Create a professional, modern and user-friendly form generation website with a teal colour theme. The website should have a clean interface, a form builder section and a separate prompt library. Make the design responsive and suitable for real-world use."

Purpose:

This prompt focused on the visual design and user experience.

Improvement:

The project was developed with a consistent teal-and-white appearance, a structured layout and separate sections for form creation and prompt selection.

Prompt 3 – Adding the Prompt Library

Prompt:

"Add a separate prompt library containing at least 25 detailed, editable sample prompts. Organize the prompts into useful categories and allow users to search, select and edit them before generating a form."

Purpose:

This prompt introduced reusable templates so users would not always need to write prompts from scratch.

Improvement:

A prompt library with more than 25 sample templates was included, along with search and category-filtering functionality.

Prompt 4 – Implementing Form Generation and Validation

Prompt:

"Implement working form generation using JavaScript. Based on the user's prompt, identify suitable fields such as name, email, date, number and dropdown inputs. Generate an interactive form with appropriate validation rules. Do not make the Generate button a placeholder."

Purpose:

This prompt focused on the main functionality of the application.

Improvement:

JavaScript was used to identify likely fields from the prompt and dynamically create forms. Validation was added for required fields and relevant input types.

Prompt 5 – Adding PDF Download and Form Sharing

Prompt:

"After a form is successfully validated, display a confirmation message and provide two options: download the completed form as a PDF and generate a shareable form link. The

PDF should contain the completed field values. The shared link should open a separate blank form that another user can fill in. Preserve the existing design and functionality."

Purpose:

This prompt introduced additional features that made the application more useful.

Improvement:

The application was updated to support browser-based PDF printing and shareable URLs containing the form structure. Recipients can open the link and complete a blank version of the form.

5. Development and Refinement Process

The application was improved through testing and troubleshooting.

The main development stages were:

1. Planning the project structure and identifying the required features.
2. Creating the website interface using HTML and CSS.
3. Implementing JavaScript functionality for form generation.
4. Developing the prompt library with reusable templates.
5. Adding validation for different input types.
6. Implementing PDF download through the browser's print functionality.
7. Creating shareable links for blank client-facing forms.
8. Testing the application and correcting issues.
9. Uploading the project to GitHub and deploying it through Vercel.

During development, instructions were refined whenever a feature did not work as expected. Existing functionality and the original website design were preserved while adding new features.

6. Final Features of FormFlow AI

The completed application includes:

- **Prompt-Based Form Generation:** Generates form fields based on the user's text description.
- **Prompt Library:** Provides reusable and editable sample prompts.
- **Dynamic Form Preview:** Displays the generated form for users.
- **Input Validation:** Checks required fields, email addresses, URLs and number limits.
- **PDF Download:** Allows completed form details to be printed or saved as a PDF.
- **Shareable Form Links:** Generates links containing the form structure.
- **Client-Facing Form:** Allows recipients to open and fill in a blank form.

7. Technical Implementation and Limitations

The current version uses HTML, CSS and JavaScript.

An important implementation detail is that form generation currently uses local JavaScript keyword-based field inference. It does not yet use a live Gemini API or another connected AI inference service.

The shareable link contains an encoded representation of the form structure. When opened, JavaScript decodes the structure and displays the blank form.

The PDF feature uses the browser's print functionality, where users can choose the Save as PDF option.

Current limitations:

- Form responses are not stored in a database.
- Responses are not automatically sent to the form creator.
- Advanced AI-based prompt understanding is not yet implemented.
- The shared link carries the form structure, not previously submitted responses.

8. Future Scope

The project can be improved further by introducing:

- Gemini API integration for more advanced prompt understanding and form generation.
- Database integration to store submitted responses.
- Automatic response collection and management.
- User authentication and individual form histories.
- More advanced customization and validation options.

9. Learning Outcomes

Through this project, I learned:

- How prompt engineering helps convert ideas into structured instructions.
- How to improve results by adding specific requirements to prompts.
- How HTML, CSS and JavaScript work together in a web application.
- How JavaScript can dynamically create and validate forms.
- How to implement browser-based PDF generation.
- How URLs can carry encoded information for sharing.

- How to identify errors, test features and improve an application through iterative development.

10. Conclusion

FormFlow AI demonstrates how prompt engineering can support the development of a practical web application.

By using clear instructions, refining requirements and improving the implementation step by step, the project evolved from an initial idea into a working form generation and validation assistant.

The development process also highlighted the importance of testing, understanding generated code and identifying the limitations of the final implementation.

The project demonstrates how effective prompting, programming knowledge and continuous improvement can be combined to build a useful application.